

RESEARCH PROJECT PORTFOLIO

UGDA-Net

Uncertainty-Guided Dual Attention Network

with Entropy-Weighted Loss and Deep Supervision for Plant Seedling Segmentation

Mohamed Ehab Mohamed

MSA University • Machine Learning Researcher • Computer Vision Specialist



Under Review • IMSA
Conference



+9.3% Dice (U-Net) • +13.2%
Dice (LinkNet)



5-Variant Ablation • 2
Backbone Architectures

01 | PROJECT OVERVIEW

UGDA-Net (Uncertainty-Guided Dual Attention Network) is a novel deep learning architecture designed to solve one of the most challenging problems in precision agriculture: accurate segmentation of plant seedlings in complex, cluttered real-world images. Standard segmentation models struggle with fine leaf structures, overlapping boundaries, soil background clutter, and inconsistent lighting — all of which degrade phenotyping accuracy in practical field conditions.

UGDA-Net addresses these challenges by combining three novel components into a unified uncertainty-aware framework: an attention module guided by feature uncertainty, an entropy-weighted loss function that focuses training on difficult boundary pixels, and deep supervision for improved gradient flow. Evaluated on a dataset of 432 high-resolution plant seedling images, UGDA-Net achieved a 9.3% Dice improvement over U-Net baseline and 13.2% over LinkNet baseline — the best results across all tested configurations.

02 | PROJECT DETAILS

Project Type	Research Paper — Novel Architecture for Uncertainty-Aware Semantic Segmentation
Domain	Computer Vision • Semantic Segmentation • Precision Agriculture • Uncertainty Estimation
Application	Automated Plant Phenotyping — Leaf Area Estimation, Growth Rate Analysis, Biomass Prediction
Dataset	432 High-Resolution RGB Plant Seedling Images with Pixel-Level Binary Masks
Backbones Tested	U-Net (ResNet34 encoder) • LinkNet (ResNet34 encoder) — both pretrained on ImageNet
Core Technologies	Python • PyTorch • U-Net • LinkNet • ResNet34 • AdamW • Mixed Precision Training
Affiliation	MSA University — Faculty of Computer Science, Giza, Egypt

03 | KEY RESULTS & METRICS

+9.3% Dice Improvement Over U-Net baseline	+13.2% Dice Improvement Over LinkNet baseline	432 Dataset Size High-res annotated images	5 × 2 Ablation Variants Configs across 2 backbones
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Full Ablation Study Results — All Variants Across Both Backbones

Model Variant	U-Net DSC	U-Net IoU	LinkNet DSC	LinkNet IoU
Baseline	0.4233	0.3197	0.3519	0.2563
Loss-Only	0.4991	0.3743	0.4714	0.3492
Attention-Only	0.4323	0.3280	0.3608	0.2563
Deep Supervision Only	0.4144	0.3107	0.4076	0.3048
UGDA-Net (Proposed)	0.5159	0.3905	0.4840	0.3557

UGDA-Net achieves the highest DSC and IoU across both backbones — outperforming all 4 ablation variants in every configuration

04 | TECHNICAL APPROACH

Three Novel Components of UGDA-Net

① Uncertainty-Guided Dual Attention (UGDA) Module The UGDA module extends traditional attention mechanisms by incorporating an uncertainty signal derived from feature map channel variance. For each encoder feature map, it computes three components simultaneously: channel attention (global average pooling + 1×1 convolutions), spatial attention (7×7 convolution + sigmoid), and an uncertainty map from channel-wise standard deviation. High variance indicates ambiguous or uncertain regions. All three are multiplied together with a learnable scalar ($\gamma = 0.1$) and combined with a residual connection — preserving original features while amplifying uncertain boundary regions.
② Entropy-Weighted Hybrid Loss Function Rather than treating all pixels equally, the entropy-weighted hybrid loss assigns higher penalties to pixels with high predictive uncertainty — primarily boundary pixels where the model is most likely to make errors. Pixel-wise entropy is computed from predicted probabilities and scaled into a weight map ($\beta = 0.3$). This weighted binary cross-entropy is combined with Dice loss ($\lambda_1 = 0.7, \lambda_2 = 0.3$). A 3-epoch warm-up phase using standard BCE stabilizes early training before switching to the full entropy-weighted hybrid loss.
③ Deep Supervision with Auxiliary Heads Auxiliary segmentation heads are placed at the two deepest encoder layers, each outputting a binary segmentation map upsampled to input resolution via bilinear interpolation. A decreasing weight scheme assigns 0.7 to the deeper layer and 0.3 to the shallower one, scaled by $\alpha = 0.05$. Deep supervision enforces multi-scale feature consistency and improves gradient flow — particularly beneficial for LinkNet whose simpler decoder

benefits more from auxiliary supervision than U-Net's skip connections.

Key Ablation Insights

- Entropy-weighted hybrid loss is the single largest contributor — validating that boundary uncertainty weighting drives the most improvement
- UGDA attention module alone yields small gains, but works synergistically with the uncertainty loss to produce the best combined result
- Deep supervision benefits LinkNet more than U-Net due to LinkNet's simpler decoder lacking built-in gradient flow mechanisms
- Full UGDA-Net is the only configuration combining all three components — producing consistent best results across both backbones
- Deterministic uncertainty estimation via channel variance achieves results without the computational overhead of Bayesian methods

05 | KEY CONTRIBUTIONS & IMPACT

First Unified Uncertainty Framework

Prior work separated uncertainty attention and uncertainty loss into independent techniques. UGDA-Net is the first architecture to unify uncertainty-guided attention, entropy-weighted loss, and deep supervision in a single end-to-end trainable framework for segmentation.

Comprehensive Ablation Study

A rigorous 5-variant ablation study across 2 backbone architectures under identical training conditions isolates and quantifies the contribution of each component — providing reproducible, actionable insights for future uncertainty-aware segmentation research.

Real-World Agricultural Impact

Accurate plant seedling segmentation directly enables automated phenotyping at scale — supporting leaf area estimation, growth rate prediction, biomass analysis, and early stress detection. UGDA-Net's improved boundary precision addresses the core bottleneck in deploying AI for precision agriculture.

Efficient Uncertainty Estimation

Unlike Bayesian methods requiring multiple stochastic forward passes, UGDA-Net's channel variance approach estimates uncertainty in a single forward pass — making it computationally practical for real-world deployment without sacrificing the benefits of uncertainty-guided training.

06 | ABOUT THE RESEARCHER

Mohamed Ehab Mohamed

Machine Learning Researcher & Engineer • Computer Vision & NLP Specialist

Education: Dual B.Sc. Computer Science — MSA University (Egypt) & University of Greenwich (UK), Expected 2026

Specialization: Computer Vision, Semantic Segmentation, NLP, LLMs, Uncertainty Estimation, Deep Learning, End-to-End ML Pipelines

Contact: me338484@gmail.com